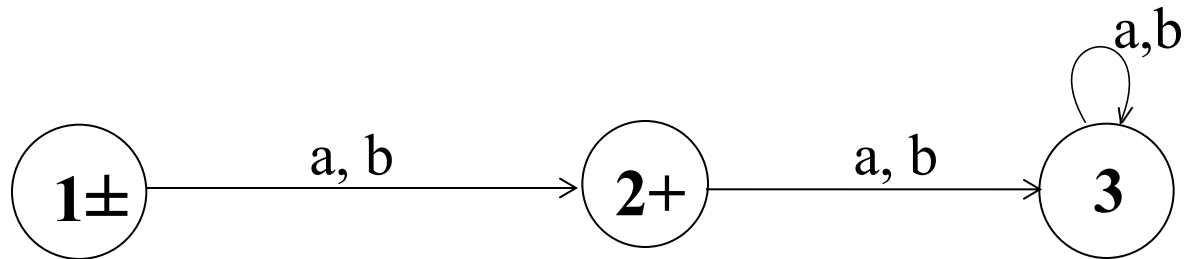


Recap lecture 18

NFA corresponding to union of
FAs, example, NFA corresponding to
concatenation of FAs, examples, NFA
corresponding to closure of an
FA, examples

Example

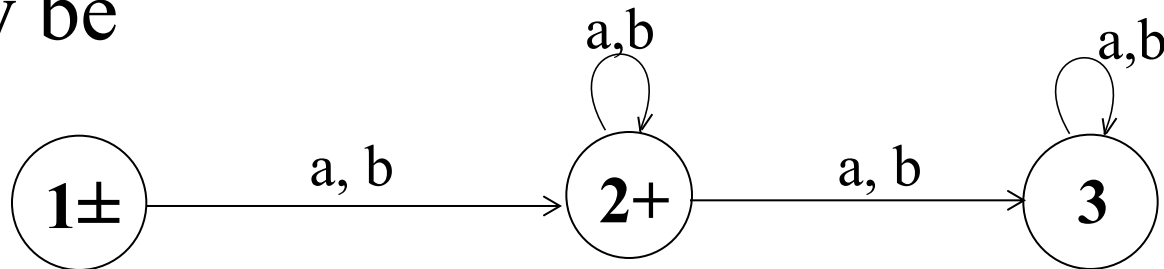
Consider the following FA



It can be observed that FA* not only accepts the Null string but every other string as well.

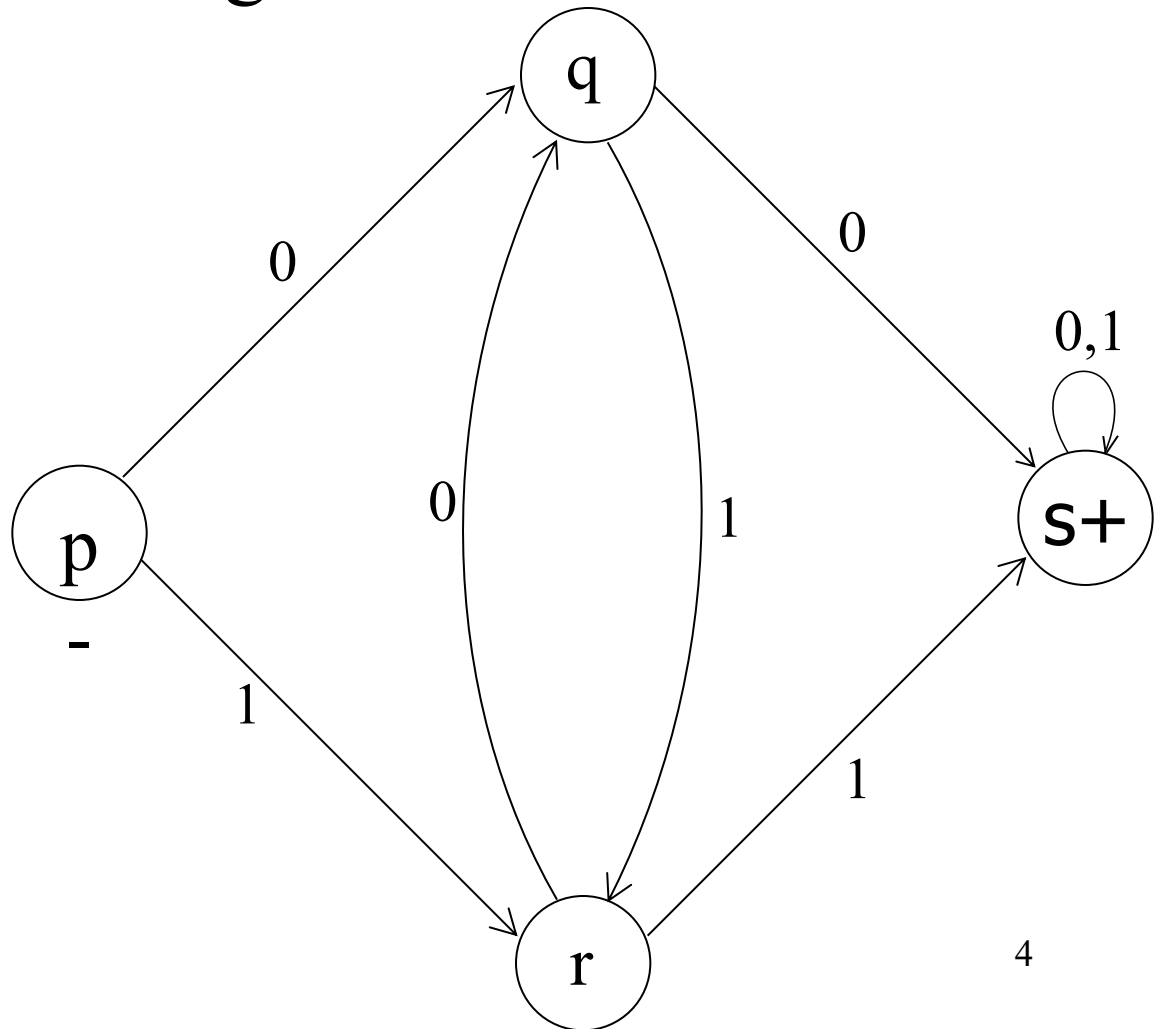
Example continued ...

Here we don't need separate initial and final state. Hence an NFA corresponding to FA^* may be



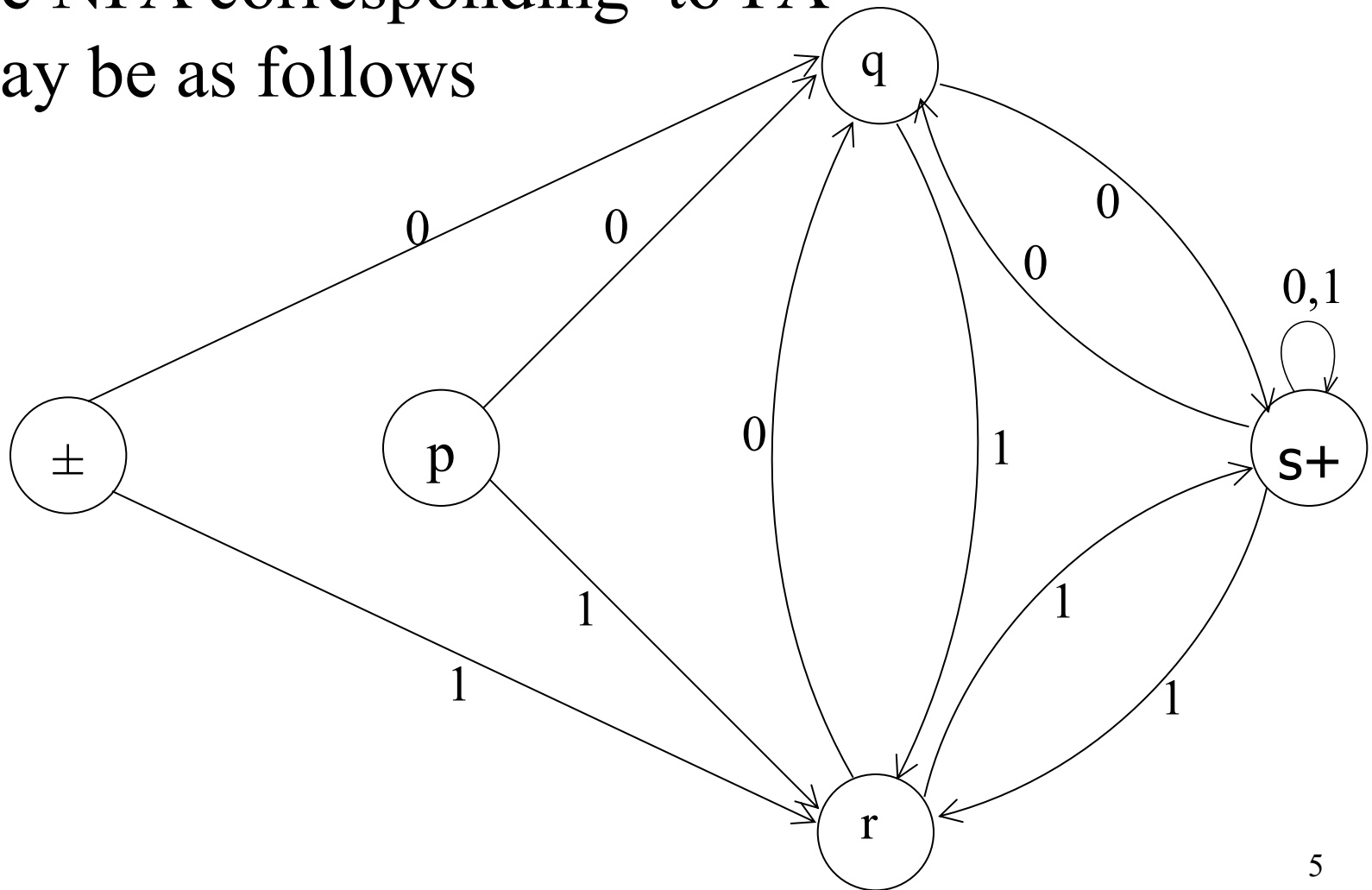
Example

- Consider the following FA



Example continued ...

the NFA corresponding to FA^*
may be as follows



Memory required to recognize a language

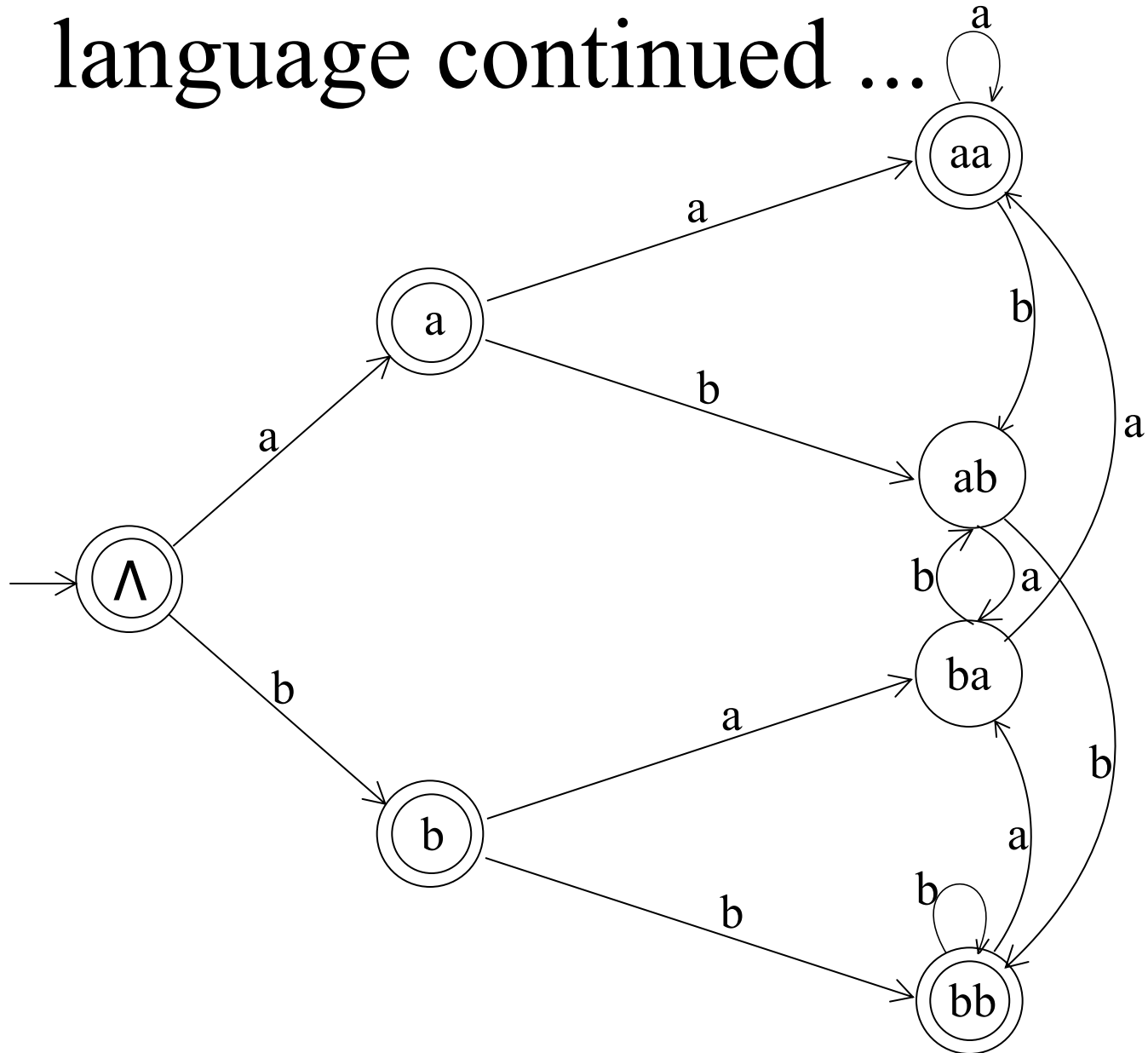
Memory required to recognize a language means to look at the machine which can recognize a language. As an FA can be considered to be a machine which is simple model of computation and every regular language is associated with certain FA, so to recognize a language there is a restriction that there is a single pass from left to right for any string to decide whether it belongs to certain language ? This helps to remember the information about the initial part of the string read so far.

Memory required to recognize a language continued ...

By this process the input string is examined and the string is decided either to be in a certain language or not.

consider $L = \{w \in \{a,b\}^* : w \text{ neither ends in } \mathbf{ab} \text{ nor in } \mathbf{ba}\}$. *i.e.* L is the language of strings, defined over $\Sigma = \{a,b\}$, consisting of Λ , a , b and strings ending in aa or bb . L may be accepted by the following FA

Memory required to recognize a language continued ...



Memory required to recognize a language continued ...

As seen in the previous FA, seven states are required to recognize the language L, while on the other hand it is very hard to recognize the language PALINDROME.

Memory required to recognize a language continued ...

As seen in the previous example of FA, seven states are required to recognize that language. Now consider another language L_3 of strings of length three or more, defined over $\Sigma = \{a,b\}$, **and the third letter from the right is a.**

Memory required to recognize a language continued ...

As discussed by Martin, there is a straight forward method to build an FA recognizing L_3 *i.e.* a distinct state for every possible substring of length less than or equal to 3. It is obvious that for each length i , $i=0,1,2,3$, of substring, the number of states are 2^i and thus total number of states required to recognize the language L_3 are $2^0+2^1+2^2+2^3 = 2^{3+1}-1=15$ (using $2^0+2^1+2^2+\dots+2^n = 2^{n+1}-1$)

Memory required to recognize a language continued ...

Remark: Let L_{20} be the language of strings of length 20 or more, defined over $\Sigma = \{a,b\}$, **and the 20th letter from the right is 1**, then following the previous method, number of states for the corresponding FA is **$2^{20+1}-1=2,097,151$** .

However, it may be noted that any portion of memory of a computer that can accommodate 21 bits can be in 2^{21} possible states *i.e.* 2^{21} possible choices for the informational content.

Distinguishable strings and Indistinguishable strings

- Two strings x and y , belonging to Σ^* , are said to be **distinguishable** w.r.t a language $L \subseteq \Sigma^*$ if there exists a string z belonging to Σ^* s.t. $xz \in L$ but $yz \notin L$ or $xz \notin L$ but $yz \in L$.
- Two strings x and y , belonging to Σ^* , are said to be **indistinguishable** with respect to a language $L \subseteq \Sigma^*$ if for every string z belonging to Σ^* , either both xz or $yz \in L$ or both don't belong to L .

Example

Let L be the language of strings, defined over $\Sigma = \{0,1\}$, **ending in 01**.

The strings 110 and 010011 are **distinguishable** w.r.t L , as there exists 1 belonging to Σ^* s.t. 1101 belongs to L but 0100111 doesn't belong to L .

But 111 and 010011 are **indistinguishable**, for 1 belonging to Σ^* s.t. both 1111 and 010011 don't belong to L *i.e.* for every z belonging to Σ^* , either both 111 z and 010011 z belong to L , or both don't belong to L .

Task

Consider the language L of strings of length three or more, defined over $\Sigma = \{0,1\}$, **ending in 011** then determine which of the following pairs are distinguishable or indistinguishable w.r.t. L .

1. 001011, 11001111
2. 01001111, 1100001111

Theorem

Statement:

If L is a language over an alphabet Σ and for integer n there are n strings from Σ^* , any two of which are distinguishable w.r.t. language L , then any FA recognizes L must have at least n states.

Note: There may not exist any FA which recognizes the given language.

Proof

Let S be set of strings, any two of which are distinguishable w.r.t. language L . Let F_1 be the FA which recognizes the language L . To prove the theorem, it is sufficient to show that any two strings under F_1 must be ended in different states *i.e.* corresponding to each string x belonging to S , F_1 ends in distinct states.

Proof continued ...

Thus if S has n strings then it is to be shown that F_1 has at least n states.

Let x and y be any two strings from S . By supposition any two strings of S are distinguishable w.r.t. L , so there exists a string z belonging to Σ^* such that only one of xz and yz belongs to L i.e. F_1 ends in a final state either for xz or yz which shows that F_1 ends in distinct states for xz and yz .

Proof continued ...

Let F_1 be end in same state for both the strings x and y , which shows that F_1 ends in same state for both xz and yz , a contradiction as x and y being distinguishable implies xz and yz are ended at distinct states of F_1 .

Hence F_1 does not end in a same state for both strings x and y , which shows that each pair of strings belonging to S ends in different states. Hence F_1 must contain at least n states.

Summing Up

NFA corresponding to Closure of FA,
Examples, Memory required to recognize
a language, Example, Distinguishing one
string from another, Example, Theorem,
Proof